# SLO, SLA, SLI & Error Budget

*+ PQRST Framework for Incident Communication*

# Objective

Equip SRE teams and engineering leaders with a shared vocabulary and operational framework to measure reliability, manage risk, and communicate incidents clearly.

### Measure

Define SLIs and SLOs to quantify real system reliability

### Manage Risk

Use Error Budgets to balance velocity with stability

### Communicate

Apply PQRST for fast, structured incident updates

# AGENDA


01 **SLI – Service Level Indicator**
Measuring actual system behavior

02 **SLO – Service Level Objective**
Setting internal reliability targets

03 **SLA – Service Level Agreement**
External contractual commitments

04 **Error Budget**
Quantifying acceptable failure

05 **How They Fit Together**
The SRE reliability pipeline

06 **PQRST Framework**
Structured incident communication

07 **Connecting SLOs + PQRST**
Data-driven incident response

08 **Key Takeaways**
Summary and next steps

# SLI

Service Level Indicator

*"How did we do?"*

An SLI is a metric and not a goal. It measures actual system behavior from the user's perspective.

*Think of SLIs as raw signals.*
*Without them, you're guessing.*

## SLI Examples

**Request Success Rate**

% of 2xx HTTP responses

**99.2%**

**Latency (p95)**

p95 response < 200ms

**187ms**

**Availability**

Uptime over 30-day window

**99.95%**

SLIs are the foundation — every SLO is built on one or more SLIs

# SLO — Service Level Objective

*"What's acceptable?"*

An SLO sets a target for an SLI. It defines the reliability level your system is expected to meet internally.

| Above SLO | Below SLO |
| --- | --- |
| ✓ Acceptable — keep shipping | ✗ Action required immediately |

## SLO Examples

### Availability SLO

99.9% of requests must succeed over 30 days

99.9%

### Latency SLO

p95 latency must be under 200ms for 99% of requests

99%

### Error Rate SLO

Error rate must stay below 0.1% monthly

99.9%

# SLA — Service Level Agreement

*"What's contractually promised?"*

An SLA is a legal/contractual commitment to customers. It typically includes a reliability guarantee and financial or contractual penalties if breached.

## SLA Example

*"We guarantee 99.5% monthly availability or you receive a 10% service credit."*

## SLO vs SLA: Why the Gap Matters

**SLO: 99.9%**

**SLA: 99.5%**

The SLO acts as a safety buffer. Teams work to the SLO so the SLA is never breached.

## Components of a Typical SLA

- **Reliability Guarantee**
- **Penalty / Credits**
- **Measurement Window**
- **Exclusions / Carve-outs**

# Error Budget — How Much Failure Is Allowed?

## Formula

Error Budget = **100% − SLO%**

*If SLO = 99.9% → Budget = 0.1%*

## Over 30 Days with 1,000,000 Requests

| | |
|---|---|
| **1,000,000** Total Requests | **99.9%** SLO Target |
| **1,000** Allowed Failures | **43.8 min** Allowed Downtime |

## How Teams Use Error Budgets

| 🚀 Budget Remaining | 🛑 Budget Exhausted | ⚡ Small Incident | 🔥 Budget Nearly Gone |
|---|---|---|---|
| → Ship features faster | → Freeze risky changes | → Low urgency if budget healthy | → Same incident = critical |

# How SLI, SLO, SLA & Error Budget Fit Together

## The Reliability Pipeline

**SLI**
Measures behavior

→

**SLO**
Sets thresholds

→

**SLA**
External commitment

→

**Error Budget**
Guides decisions

## In Practice — 5 Steps

1. SLIs measure real system behavior
2. SLOs define acceptable performance thresholds
3. SLOs are set stricter than SLAs to create a buffer
4. Error Budgets quantify how much deviation is allowed
5. Teams use Error Budget to guide deployments, risk-taking & incident response

# PQRST Framework for Incident Communication

*During an incident, confusion is the enemy. PQRST ensures every update is structured, fast, and unambiguous.*

## P Provocation

*"What triggered this?"*

What signal alerted you? E.g. "Spike in HTTP 500 errors from checkout service."

## Q Quality

*"What's actually broken?"*

User-facing impact. E.g. "Customers are unable to complete purchases."

## R Region

*"Where is this happening?"*

Scope the issue. E.g. "Only us-west-2 affected" or "All mobile clients."

## S Severity

*"How bad is it?"*

Quantify impact. E.g. "10% of requests failing" or "100% outage for new sessions."

## T Time

*"When did it start?"*

Establish timeline. E.g. "First alert at 09:32 PT; errors rising since 09:28."

# PQRST In Action — SLO-Aware Incident Update

## ❌ Without PQRST

*"System is having issues"*

## ✅ With PQRST

*"500 errors at 12% in us-west-2 since 09:28, affecting checkout"*

### Full SLO-Aware PQRST Update:

| | | |
|---|---|---|
| P | **Provocation:** | Spike in 500 errors from the checkout service |
| Q | **Quality:** | Customers cannot complete purchases |
| R | **Region:** | All regions affected |
| S | **Severity:** | 8% failure rate — consuming 50% of monthly error budget in 1 hour |
| T | **Time:** | Started at 09:28 PT following deployment |

# Connecting SLOs + PQRST — Data-Driven Incident Response

## SLO / Error Budget — Defines Urgency

- Are we breaching an SLO?
- How fast are we burning the error budget?
- Is the SLA at risk?
- 2% failure = low urgency if budget is healthy
- 2% failure = critical if budget nearly exhausted

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## PQRST — Communicates Urgency

- Engineers understand technical impact
- Leadership understands business risk
- Everyone understands time pressure
- Structured → no ambiguity, faster triage
- Shared mental model across all stakeholders

*Together: chaotic guesswork → data-driven decision-making + precise communication*

# Key Takeaways

1. SLIs are your raw signal. Without measurement, reliability is guesswork.
2. SLOs set the engineering line between acceptable and action-required.
3. SLAs are contractual, always set SLOs stricter to maintain a safety buffer.
4. Error Budgets balance velocity and reliability. They make risk decisions objective.
5. PQRST transforms 'system issues' into clear, structured, actionable updates.
6. SLO + PQRST together = data-driven urgency + precise stakeholder communication.